

ducted. They can range in size from a small appliance-size equipment, which is well-suited for small-scale distributed hydrogen production, to a large-scale central production facility linked to renewable or other non-greenhouse-gas-emitting forms of electricity production. Some of the different processes used are:

Polymer electrolyte membrane (PEM) electrolyzers. PEM electrolyzers (shown in Fig. 2), use a solid polymer electrolyte (SPE). Water reacts at the anode to form oxygen and positively charged hydrogen ions (protons). The electrons flow through the external circuit and the hydrogen

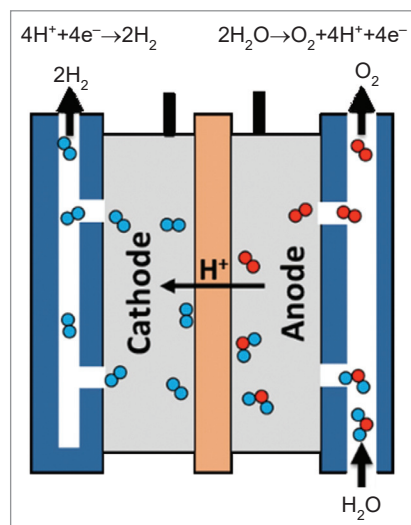


Fig. 2: Polymer electrolyte membrane (PEM) electrolyser (Source: www.energy.gov/eere)

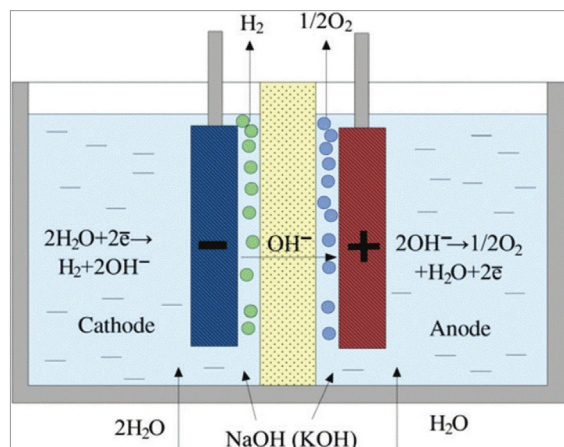


Fig. 3: Alkaline electrolyser (Source: <https://www.researchgate.net>)

ions selectively drift across the PEM to the cathode. At the cathode, they combine with the electrons from the external circuit to form hydrogen gas.

Alkaline electrolyzers. Alkaline electrolyzers, (Fig. 3) use liquid alkaline, such as potassium hydroxide or sodium hydroxide, as the electrolyte solutions. The electrodes are separated by a diaphragm, which separates oxygen and hydrogen and transports hydroxide ions (OH^-) from the cathode to the anode generating hydrogen on the cathode. At the cathode, the water molecules are reduced by electrons to form hydrogen and negatively charged (OH^-) ions. At the anode the (OH^-) ions are oxidised to oxygen and water while releasing electrons. Alkaline electrolyzers provide higher efficiency than the PEM electrolyzers.

Solid oxide electrolyzers (SOEs). SOEs operate at a higher temperature, $700^\circ\text{--}800^\circ\text{C}$, as compared to the PEM electrolyzers which operate at $70^\circ\text{--}90^\circ\text{C}$ and alkaline electrolyzers which operate at less than 100°C . SOEs, Fig. 4, use a solid ceramic material as the electrolyte, which selectively conducts the negatively charged oxygen ions (O^{2-}). The oxygen ions pass through the solid ceramic membrane and react at the anode to form oxygen and generate electrons for the external circuit.

Hot steam produced at the cathode combines with the electrons from the external circuit to form hydrogen. SOEs using proton-conducting ceramic electrolytes are also available, which operate at comparatively lower temperatures of $500^\circ\text{--}600^\circ\text{C}$. The SOE technology is showing a great promise.

Solar thermochemical water splitters. Solar thermochemical water

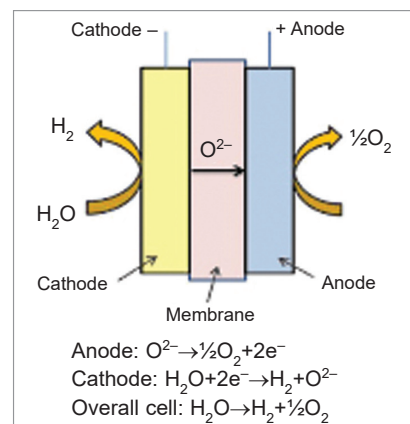


Fig. 4: Solid Oxide Electrolyser (SOE) (Source: <https://www.researchgate.net>)

splitting processes (STCH) use high-temperature ($500^\circ\text{--}2,000^\circ\text{C}$) to drive a series of chemical reactions that produce hydrogen. The high temperatures are generated by concentrating sunlight onto a reactor tower using a field of mirror 'heliostats,' as shown in Fig. 5. Waste heat from the advanced nuclear reactors can also be used to generate high temperatures. The chemicals used in the process are reused within each cycle, creating a closed loop that consumes only water and produces hydrogen and oxygen.

Photoelectrochemical (PEC) water splitting process. In PEC processes, hydrogen is produced from water using specialised semiconductors called photoelectrochemical materials, which use sunlight energy to dissociate water molecules into hydrogen and oxygen. The semiconductor materials are immersed in a water-based electrolyte, where sunlight energises the water-splitting process. The PEC reactors are constructed in panel form (similar to the photovoltaic panels) as electrode systems or as slurry-based particle systems. The panel systems are most popular, as the semiconductor materials used in the process are similar to those used in photovoltaic solar electricity generation. This process offers higher conversion efficiency as it operates at comparatively lower temperatures.